

Given: Due:	Further Maths (Core Pure) HW 22				Name:	
Chapter 8: Proof by Induction						
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.						
91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)	
Useful resources: Your notes from recent lessons & your text book						
Questions - do these on lined paper						
Q1. Prove by induction that, for $n \in \mathbb{Z}^+$, $f(n) = 8^n - 2^n$ is divisible by 6 (6)						
Q2. (a) Prove by induction, that for $n \in \mathbb{Z}^+$, $\sum_{r=1}^n r(2r-1) = \frac{1}{6}n(n+1)(4n-1)$ (6) (b) Hence, show that $\sum_{r=n+1}^{3n} r(2r-1) = \frac{1}{3}n(an^2 + bn + c)$ where a , b and c are integers to be found. (4)						
Q3. (i) Prove by induction that for $n \in \mathbb{Z}^+$ $\begin{pmatrix} 5 & -8 \\ 2 & -3 \end{pmatrix}^n = \begin{pmatrix} 4n+1 & -8n \\ 2n & 1-4n \end{pmatrix}$ (6) (ii) Prove by induction that for $n \in \mathbb{Z}^+$ $f(n) = 4^{n+1} + 5^{2n-1}$ is divisible by 21 (6)						
Q4. (i) Prove by induction that, for $n \in \mathbb{Z}^+$. $\begin{pmatrix} 1 & 0 \\ -1 & 5 \end{pmatrix}^n = \begin{pmatrix} 1 & 0 \\ -\frac{1}{4}(5^n - 1) & 5^n \end{pmatrix}$ (6) (ii) Prove by induction that, for $n \in \mathbb{Z}^+$. $\sum_{r=1}^n (2r-1)^2 = \frac{1}{3}n(4n^2 - 1)$ (6)						
END OF HOMEWORK						
Total = 40 marks						
Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.						
File this question paper in your file, ready for revision and file inspection.						